5.5 Optimization

Name _____ Period ____ Date _____

AP Calculus 5.5 (30-minute plan) · Student sheet

I can…

- write the quantity to maximize or minimize as a function of one variable.
- find and justify the maximum or minimum.
- solve area, volume, number, and distance optimization problems.

Vocabulary · fill in the missing word

Objective function — the quantity to be _____ or minimized, written as a function

Constraint — an equation that relates the _____, used to write the objective in one variable

Part 1 · Notes and examples (fill in each note, then work the example)

Step 1: draw a picture and name the variables; write the _____ function.

Step 2: use the _____ to write the objective function in one variable, and state its domain.

Example 1

Find the largest area of a rectangle with perimeter 40 ft.

Step 3: find the critical numbers, then justify with the endpoints or the _____ Derivative Test.

Example 2

A farmer has 400 ft of fence for a rectangle along a river (no fence on the river side). Find the largest area.

Example 3

An open box is made from a 12-in square by cutting squares of side x from the corners. Find the largest volume.

Part 2 · Partner practice

Partner practice

(AP® Practice) A rectangle has a perimeter of 24. Its maximum possible area is

(A) 36 (B) 32 (C) 144 (D) 24